Examining the nurse-to-patient ratio in Taiwan urban-rural divide

Using the years of the onset, severity, and mitigation of COVID-19 as a reference—

Group 4

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Motivation and Objective

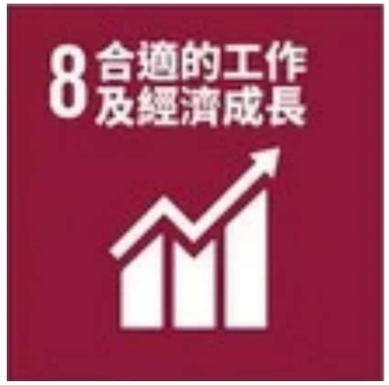
motivation

1 終結貧窮
2 消除飢餓
3 健康與福祉
4 優質教育
5 性別平權
6 淨水及衛生
7 可負擔的潔淨能源
8 合適的工作及經濟成長
9 工業化、創新及基礎建設
10 減少不平等
11 永續城鄉
12 責任消費及生產
13 氣候行動
14 保育海洋生態
15 保育陸域生態
16 和平、正義及健全制度
17 多元夥伴關係
THE GLOBAL GOALS
For Sustainable Development

motivation

SDG Goal 8: Foster inclusive and sustainable economic growth to ensure that all individuals have access to quality employment opportunities.

8.8 Safeguard the rights of workers and establish a safe and secure working environment for all employees, including foreign migrant workers, with particular attention to female migrant workers and those involved in hazardous occupations.



motivation

SDG Goal 10: Mitigate both domestic and international inequalities.

10.1 By 2030, the income of the lowest 40% of the population is expected to rise gradually at a rate surpassing the national average.

各縣市每平方公里醫師數比較

名次	縣市	醫師數	名次	縣市	醫師數
1	台北市	41.82	12	連江縣	0.69
2	嘉義市	15.59	13	雲林縣	0.69
3	新竹市	8.72	14	金門縣	0.59
4	基隆市	5.77	15	新竹縣	0.46
5	桃園市	3.54	16	嘉義縣	0.45
6	台中市	3.03	17	屏東縣	0.42
7	新北市	3.02	18	宜蘭縣	0.33
8	高雄市	2.35	19	苗栗縣	0.32
9	彰化縣	2.02	20	花蓮縣	0.19
10	台南市	1.81	21	南投縣	0.16
11	澎湖縣	0.99	22	台東縣	0.09

資料來源 / 醫師公會全聯會　製表 / ETtoday新聞雲

Purpose

- Resources are distributed unevenly across Taiwan's counties and cities, particularly highlighting a pronounced urban-rural divide.
- Consequently, we aim to investigate the correlation between the nurse-to-patient ratio and this urban-rural disparity.
- We selected two cities from northern, central, and southern Taiwan, distinguishing between urban and rural areas based on their classification as special municipalities.

The six major cities are Taipei, Taichung, and Tainan (cities), along with Yilan, Nantou, and Pingtung (townships). Collectively, these six cities and counties will be designated as the Six Major Cities.

city

1. Taipei
2. Taichung
3. Tainan

Township

4. Yilan
5. Nantou
6. Pingtung

Purpose

- In addition to investigating the correlation between the nurse-to-patient ratio and the urban-rural disparity, this study assesses whether the already critically high nurse-to-patient ratio in Taiwan has resulted in increasingly demanding and burdensome work for nurses due to the COVID-19 pandemic.
- This analysis explores the relationship between the nurse-to-patient ratio and COVID-19, as well as the connection between the nurse-to-patient ratio and the urban-rural gap.

The areas I wish to explore are as follows:

1. understand whether the nurse-to-patient ratio is associated with the urban-rural disparity.
2. comprehend the correlation between the nurse-to-patient ratio and the three years of the COVID-19 pandemic (2019, 2020, 2021).

- The onset of the epidemic
- The epidemic is critical
- The epidemic has subsided

Formulation of Hypotheses

Assume 1

Objective 1: Over the three years of the COVID-19 pandemic, was the average nurse-to-patient ratio associated with the urban-rural disparity?

H0 = The disparity in the average nurse-to-patient ratio between urban and rural areas is ≤ 1 ($p \geq \alpha$, indicating no significant difference)

H1 = The disparity in the average nurse-to-patient ratio between urban and rural areas exceeds 1 ($p<\alpha$ signifies a statistically significant difference).

- The T-test (mean test) can be employed to assess whether the disparity in the average nurse-to-disease ratio between urban and rural areas is significantly greater than 1.
- A significance level of $\alpha = 0.05$ was established to ascertain whether to reject H0.

Assume 2

Objective 2: Over the three years of the COVID-19 pandemic, what was the average nurse-to-patient ratio in the six major cities impacted by the

$$\begin{cases} H0 = \text{The average patient-to-disease ratio remains unaffected by the severity of COVID-19 } (\mu A=\mu B=\mu C). \\ \text{The average patient-to-disease ratio is influenced by the severity of COVID-19 } (\mu A\neq\mu B\neq\mu C). \end{cases}$$

- Given that the severity of COVID-19 is categorized into three levels, ANOVA (analysis of variance) can be employed to assess whether significant differences exist in the mean disease-to-patient ratio across these levels.
- Additional post-hoc testing may be performed to identify the groups that exhibit differences.

References and Sources

Nursing-to-patient ratio equation

Nurse-to-patient ratio = (Number of acute general ward beds * Bed occupancy rate * 3) / Average number of nursing personnel per shift (3 shifts)

- Acute general hospital beds: Tallied based on the quantity of "acute general hospital beds" registered and operational with the local health bureau.
- Bed occupancy rate: The calculation is derived from the bed occupancy rate for the current month.
  a. Monthly bed occupancy rate = (Total inpatient days for this bed type during the month) ÷ (Number of beds of this type × Number of days in the month) × 100 (%)
  b. Hospitalization person-days: total number of inpatients who have completed admission procedures each day during the current month. The calculation method includes inpatients while excluding outpatients; each inpatient who is admitted and discharged on the same day is counted as one hospitalization person-day.
- Number of nursing personnel: Head nurses and nursing staff, excluding specialized nurses and intern nurses.

Nursing ratio data source

全日平均護病比資訊公開

健保諮詢免付費專線：0800-030-598-衛生福利部中央健康保險署,NHI-健保資料站-健保統計專區-醫務管理-全日平均護病比資訊公開

衛生福利部中央健康保險署 / Nov 17, 2023

首頁 / 健保資料站 / 健保統計專區 / 醫務管理 / 全日平均護病比資訊公開

全日平均護病比資訊公開

健保諮詢免付費專線：0800-030-598

檔案下載

113年3月各醫院護病比明細(113.05.29新增)	ods 86 KB	pdf 531 KB
113年2月各醫院護病比明細(113.05.15新增)	ods 86 KB	pdf 532 KB
113年1月各醫院護病比明細(113.03.27新增)	ods 35 KB	pdf 530 KB
112年12月各醫院護病比明細(113.02.22新增)	ods 49 KB	pdf 531 KB
112年11月各醫院護病比明細(113.02.22修訂)	ods 64 KB	pdf 533 KB
112年10月各醫院護病比明細(112.12.22新增)	ods 49 KB	pdf 524 KB
112年9月各醫院護病比明細(112.11.17新增)	ods 55 KB	pdf 528 KB
112年8月各醫院護病比明細(112.10.31新增)	ods 36 KB	pdf 531 KB
112年7月各醫院護病比明細(112.09.15新增)	ods 36 KB	pdf 529 KB
112年6月各醫院護病比明細(112.08.18新增)	ods 54 KB	pdf 591 KB
112年5月各醫院護病比明細(112.09.13修訂)	ods 49 KB	pdf 536 KB
112年4月各醫院護病比明細(112.09.13修訂)	ods 50 KB	pdf 533 KB
112年3月各醫院護病比明細(112.09.13修訂)	ods 54 KB	pdf 538 KB

Nursing ratio data source

111年12月各醫院護病比明細(112.03.06新增)

ods 48 KB

pdf 535 KB

111年12月全民健康保險特約醫院於VPN登錄之各月份全日平均護病比
(依醫事機構代號排序)

特約類別	精神專科醫院註記	機構代號	機構名稱	院區	急性一般病床數[1]	占床率	全日平均三班護理人員數	護病比[2]
醫學中心		0401180014	台大醫院		1,723	87.8%	649.5	7
醫學中心		0421040011	成大醫院		946	89.0%	353.3	7.1
醫學中心		0501110514	三軍總醫院		1,368	74.3%	418.7	7.3
醫學中心		0601160016	臺北榮總		2,429	76.9%	835.1	6.7
醫學中心		0602030026	高雄榮總		1,019	77.9%	289.7	8.2
醫學中心		0617060018	臺中榮總		1,009	90.2%	351.3	7.8
醫學中心		1101010012	台北長庚		119	62.6%	32	7
醫學中心		1101020018	國泰醫院		590	59.6%	140.5	7.5
醫學中心		1101100011	台北馬偕		682	73.9%	224	6.7
醫學中心		1101150011	新光醫院		543	78.6%	178.5	7.2
醫學中心		1131010011	亞東醫院		866	85.8%	283	7.9
醫學中心		1131100010	馬偕淡水		743	72.2%	228.5	7
醫學中心		1132070011	林口長庚		2,554	82.0%	850.6	7.4
醫學中心		1137010024	彰基		836	85.6%	286.2	7.5
醫學中心		1141310019	奇美醫院		842	85.6%	301.5	7.2
醫學中心		1142100017	高雄長庚		1,487	84.5%	501.9	7.5
醫學中心		1145010010	慈濟醫院		535	89.2%	172.3	8.3
醫學中心		1301200010	萬芳醫院		538	78.9%	164.8	7.7
醫學中心		1302050014	高雄醫學大		1,241	89.9%	433.6	7.7
醫學中心		1317040011	中山附醫		826	78.5%	263	7.4
醫學中心		1317050017	中國附醫		1,386	83.6%	414	8.4
醫中兒醫		0401180023	台大兒醫		133	71.2%	43.5	6.5
醫中兒醫		1101100020	馬偕兒童醫		82	82.9%	32	6.4
醫中兒醫		1137010042	彰基兒童醫		113	63.7%	33.7	6.4
醫中兒醫		1303260014	中國兒童醫		120	52.2%	25.6	7.3
區域醫院		0101090517	臺北市聯醫	仁愛	449	60.3%	90	9
區域醫院		0101090517	臺北市聯醫	中興	320	54.7%	68.3	7.7
區域醫院		0101090517	臺北市聯醫	忠孝	269	58.0%	53.5	8.7
區域醫院		0101090517	臺北市聯醫	陽明	333	61.1%	66	9.2

Compilation of nursing ratio data

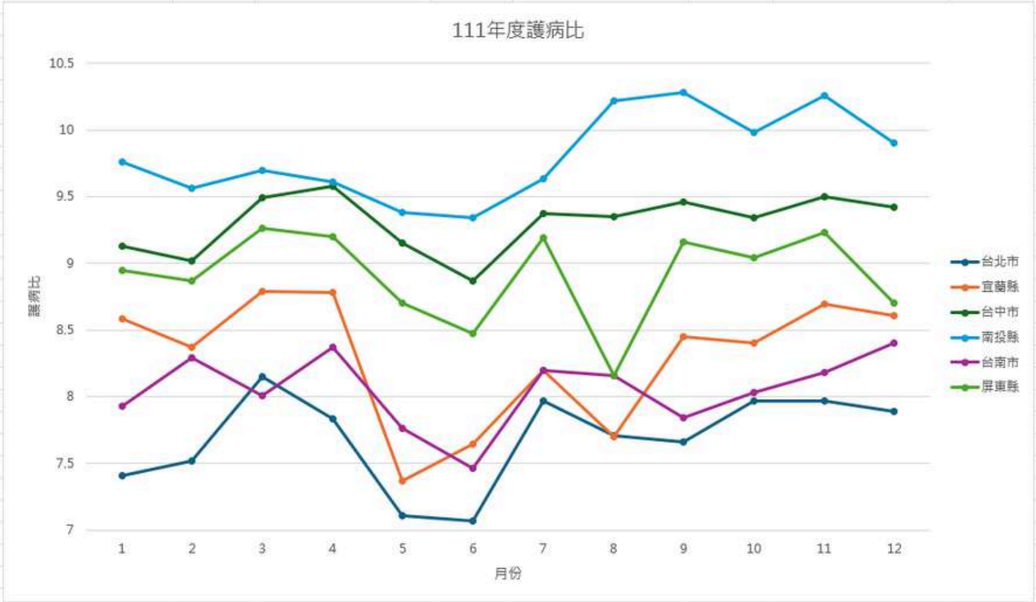
縣市	機構名稱	急性一般病床數	占床率	全日平均三班護理人員數	護病比
台北市	台大醫院	1,723	87.8%	649.5	7
台北市	三軍總醫院	1,368	74.3%	418.7	7.3
台北市	臺北榮總	2,429	76.9%	835.1	6.7
台北市	台北長庚	119	62.6%	32	7
台北市	國泰醫院	590	59.6%	140.5	7.5
台北市	台北馬偕	682	73.9%	224	6.7
台北市	新光醫院	543	78.6%	178.5	7.2
台北市	萬芳醫院	538	78.9%	164.8	7.7
台北市	台大兒醫	133	71.2%	43.5	6.5
台北市	馬偕兒童醫	82	82.9%	32	6.4
台北市	臺北市聯醫仁愛	449	60.3%	90	9
台北市	臺北市聯醫中興	320	54.7%	68.3	7.7
台北市	臺北市聯醫忠孝	269	58.0%	53.5	8.7
台北市	臺北市聯醫陽明	333	61.1%	66	9.2
台北市	臺北市聯醫松德	371	66.1%	88.5	8.3
台北市 111-12平均					7.893333
宜蘭縣	陽大醫院	400	80.7%	96.2	10.1
宜蘭縣	羅東博愛	689	66.2%	138.4	9.9
宜蘭縣	羅東聖母	444	57.5%	76.4	10
宜蘭縣	為恭醫院	399	67.6%	75.1	10.8
宜蘭縣	蘇澳榮民醫	98	71.1%	19	11
宜蘭縣	員山榮民醫	190	63.1%	34	10.6
宜蘭縣	海天醫院	40	69.0%	8	10.3
宜蘭縣	宜蘭仁愛醫	54	47.5%	13	5.9
宜蘭縣	杏和醫院	20	8.9%	3	1.8
宜蘭縣	宜蘭員山	20	57.4%	5	6.9
宜蘭縣	杏和醫院	48	85.6%	16.5	7.5
宜蘭縣 111-12平均					8.618182
台中市	臺中榮總	1,009	90.2%	351.3	7.8
台中市	中山附醫	826	78.5%	263	7.4

6都各平均值 | 城縣平均值 | 111-1 | 111-2 | 111-3 | 111-4 | 111-5 | 111-6 | 111-7 | 111-8 | 111-9 | 111-10 | 111-11 | 111-12

Compilation of nursing ratio data

Summary table of monthly average values for six selected counties and municipalities.

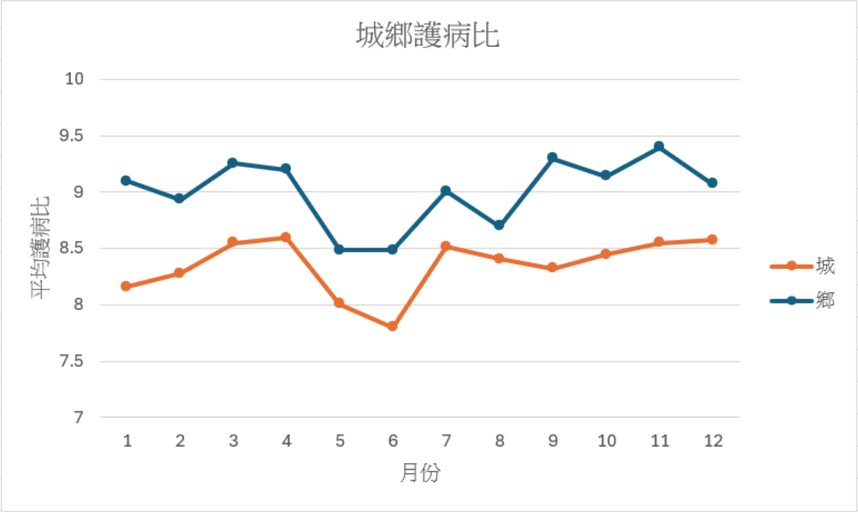
	A	B	C	D	E	F	G	H	I	J	K	L
1	台北市 111-01平均	7.41	宜蘭縣 111-01平均	8.58	台中市 111-01平均	9.13	南投縣 111-01平均	9.76	台南市 111-01平均	7.93	屏東縣 111-01平均	8.95
2	台北市 111-02平均	7.52	宜蘭縣 111-02平均	8.37	台中市 111-02平均	9.02	南投縣 111-02平均	9.56	台南市 111-02平均	8.29	屏東縣 111-02平均	8.87
3	台北市 111-03平均	8.15	宜蘭縣 111-03平均	8.79	台中市 111-03平均	9.49	南投縣 111-03平均	9.7	台南市 111-03平均	8.01	屏東縣 111-03平均	9.26
4	台北市 111-04平均	7.83	宜蘭縣 111-04平均	8.78	台中市 111-04平均	9.58	南投縣 111-04平均	9.61	台南市 111-04平均	8.37	屏東縣 111-04平均	9.2
5	台北市 111-05平均	7.11	宜蘭縣 111-05平均	7.37	台中市 111-05平均	9.15	南投縣 111-05平均	9.38	台南市 111-05平均	7.76	屏東縣 111-05平均	8.7
6	台北市 111-06平均	7.07	宜蘭縣 111-06平均	7.64	台中市 111-06平均	8.87	南投縣 111-06平均	9.34	台南市 111-06平均	7.46	屏東縣 111-06平均	8.47
7	台北市 111-07平均	7.97	宜蘭縣 111-07平均	8.2	台中市 111-07平均	9.37	南投縣 111-07平均	9.63	台南市 111-07平均	8.2	屏東縣 111-07平均	9.19
8	台北市 111-08平均	7.71	宜蘭縣 111-08平均	7.7	台中市 111-08平均	9.35	南投縣 111-08平均	10.22	台南市 111-08平均	8.16	屏東縣 111-08平均	8.16
9	台北市 111-09平均	7.66	宜蘭縣 111-09平均	8.45	台中市 111-09平均	9.46	南投縣 111-09平均	10.28	台南市 111-09平均	7.84	屏東縣 111-09平均	9.16
10	台北市 111-10平均	7.97	宜蘭縣 111-10平均	8.4	台中市 111-10平均	9.34	南投縣 111-10平均	9.98	台南市 111-10平均	8.03	屏東縣 111-10平均	9.04
11	台北市 111-11平均	7.97	宜蘭縣 111-11平均	8.69	台中市 111-11平均	9.5	南投縣 111-11平均	10.26	台南市 111-11平均	8.18	屏東縣 111-11平均	9.23
12	台北市 111-12平均	7.89	宜蘭縣 111-12平均	8.61	台中市 111-12平均	9.42	南投縣 111-12平均	9.9	台南市 111-12平均	8.4	屏東縣 111-12平均	8.7
13												
14	台北市 111 平均	7.68833	宜蘭縣 111 平均	8.29833								
15	台中市 111 平均	9.30667	南投縣 111 平均	9.80167								
16	台南市 111 平均	8.0525	屏東縣 111 平均	8.91083								
17		8.34917		9.00361								



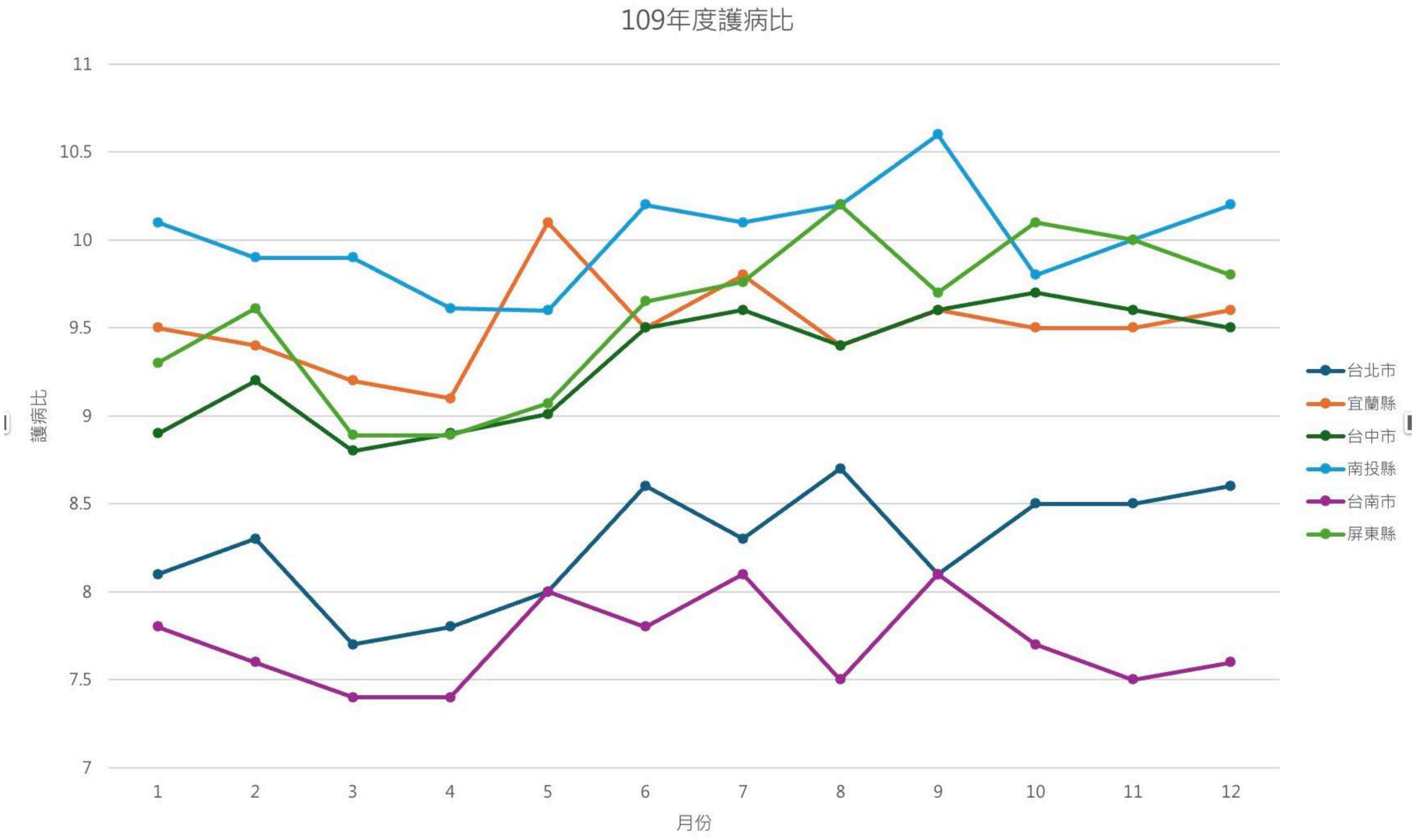
Compilation of nursing ratio data

Urban-rural average analysis

月份	台北	台中	台南	平均		宜蘭	南投	屏東	平均
1	7.41	9.13	7.93	8.15667		8.58	9.76	8.95	9.09667
2	7.52	9.02	8.29	8.27667		8.37	9.56	8.87	8.93333
3	8.15	9.49	8.01	8.55		8.79	9.7	9.26	9.25
4	7.83	9.58	8.37	8.59333		8.78	9.61	9.2	9.19667
5	7.11	9.15	7.76	8.00667		7.37	9.38	8.7	8.48333
6	7.07	8.87	7.46	7.8		7.64	9.34	8.47	8.48333
7	7.97	9.37	8.2	8.51333		8.2	9.63	9.19	9.00667
8	7.71	9.35	8.16	8.40667		7.7	10.22	8.16	8.69333
9	7.66	9.46	7.84	8.32		8.45	10.28	9.16	9.29667
10	7.97	9.34	8.03	8.44667		8.4	9.98	9.04	9.14
11	7.97	9.5	8.18	8.55		8.69	10.26	9.23	9.39333
12	7.89	9.42	8.4	8.57		8.61	9.9	8.7	9.07



Statistical Analysis


109年度護病比
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護病比
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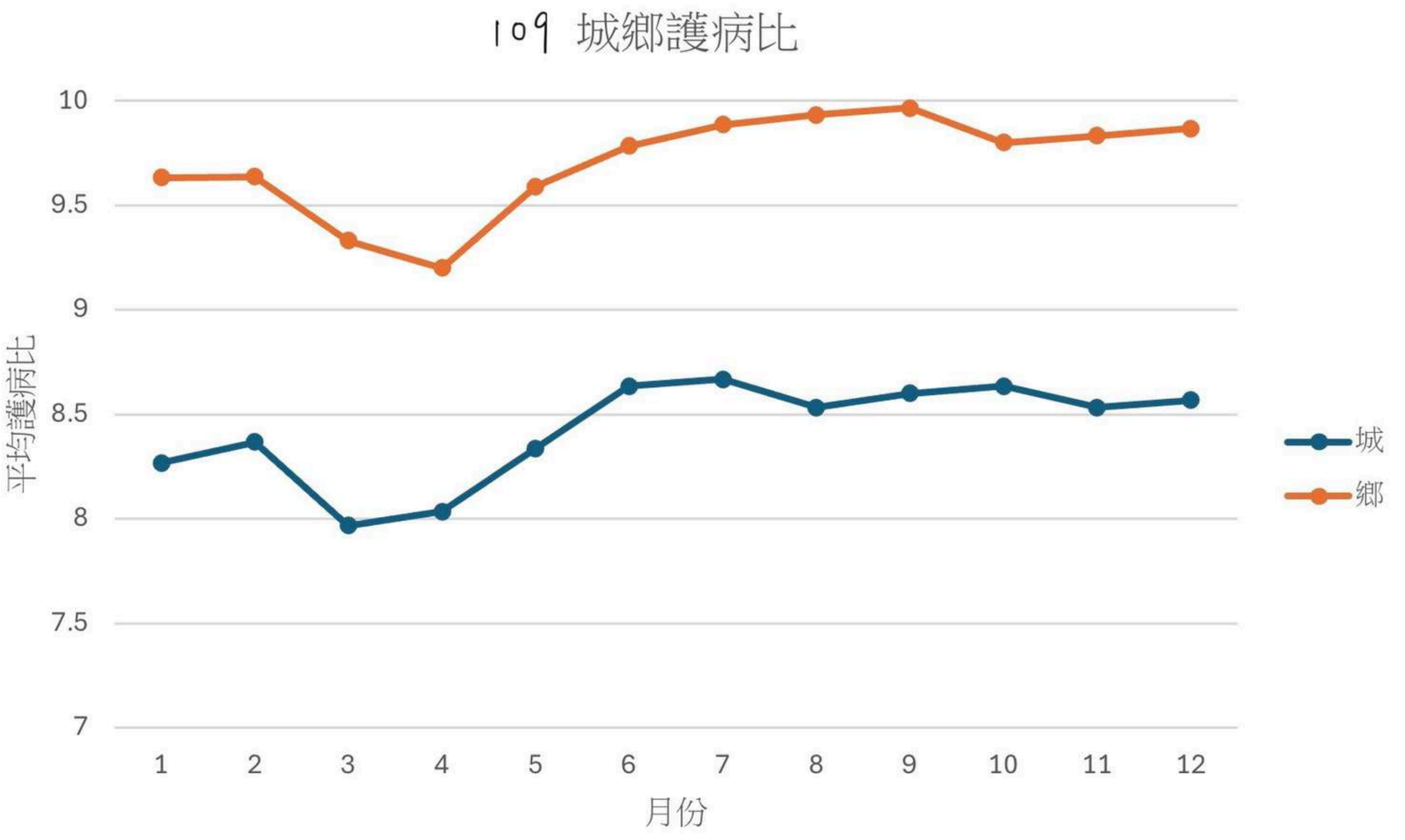
109 城鄉護病比
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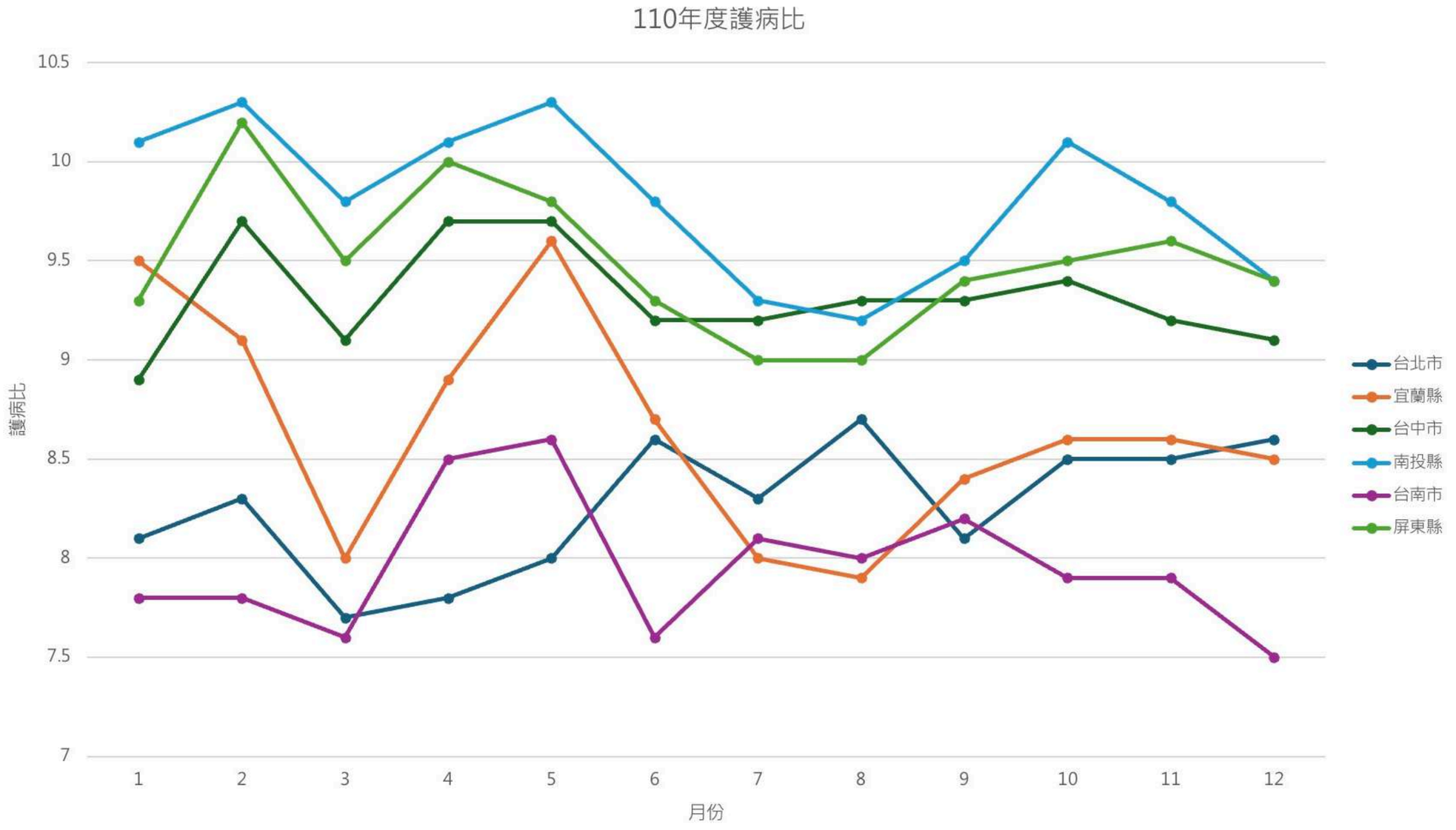
110年度護病比
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9.5
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8.5
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護病比
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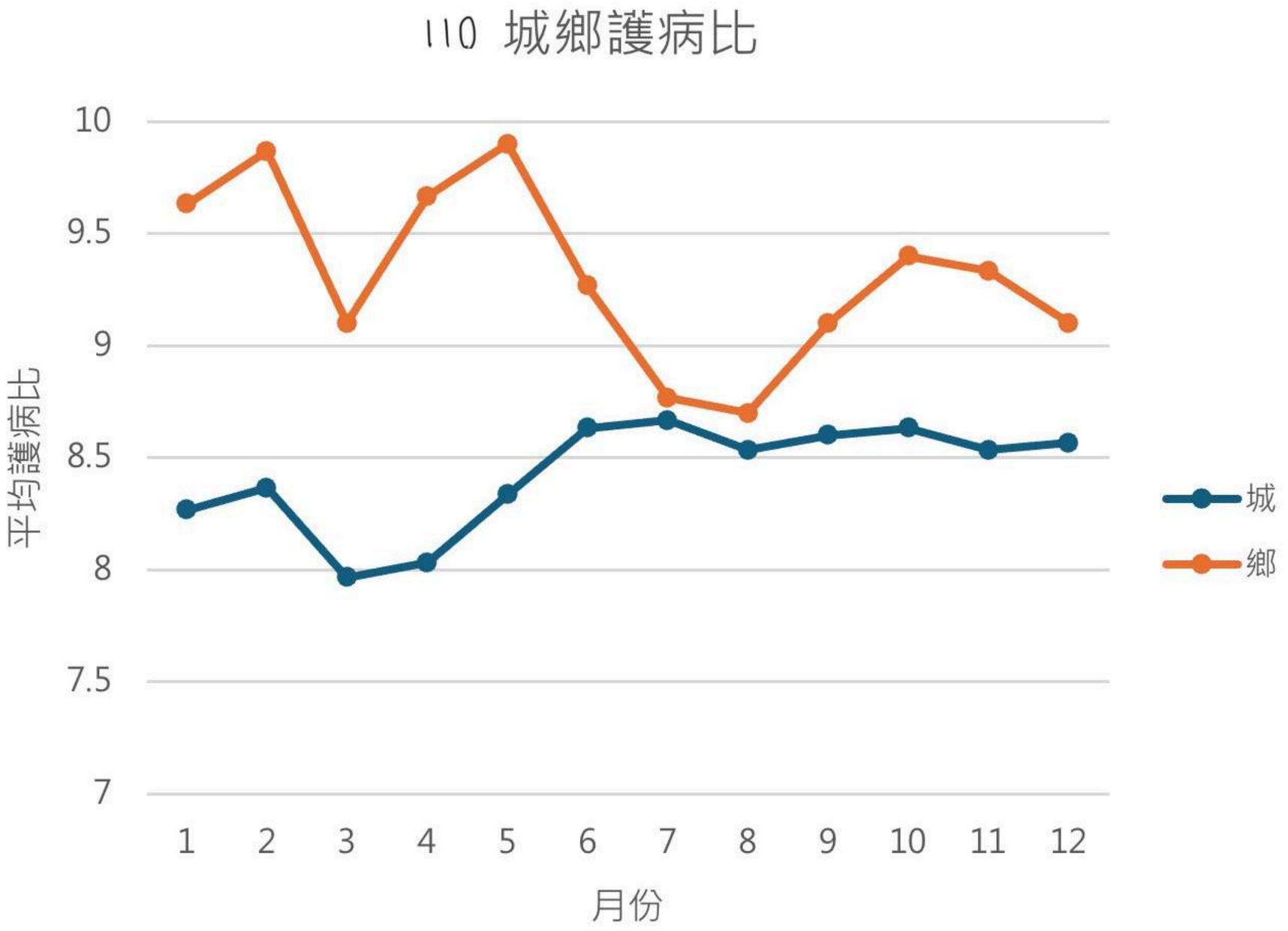
月份
台北市
宜蘭縣
台中市
南投縣
台南市
屏東縣


110 城鄉護病比
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平均護病比
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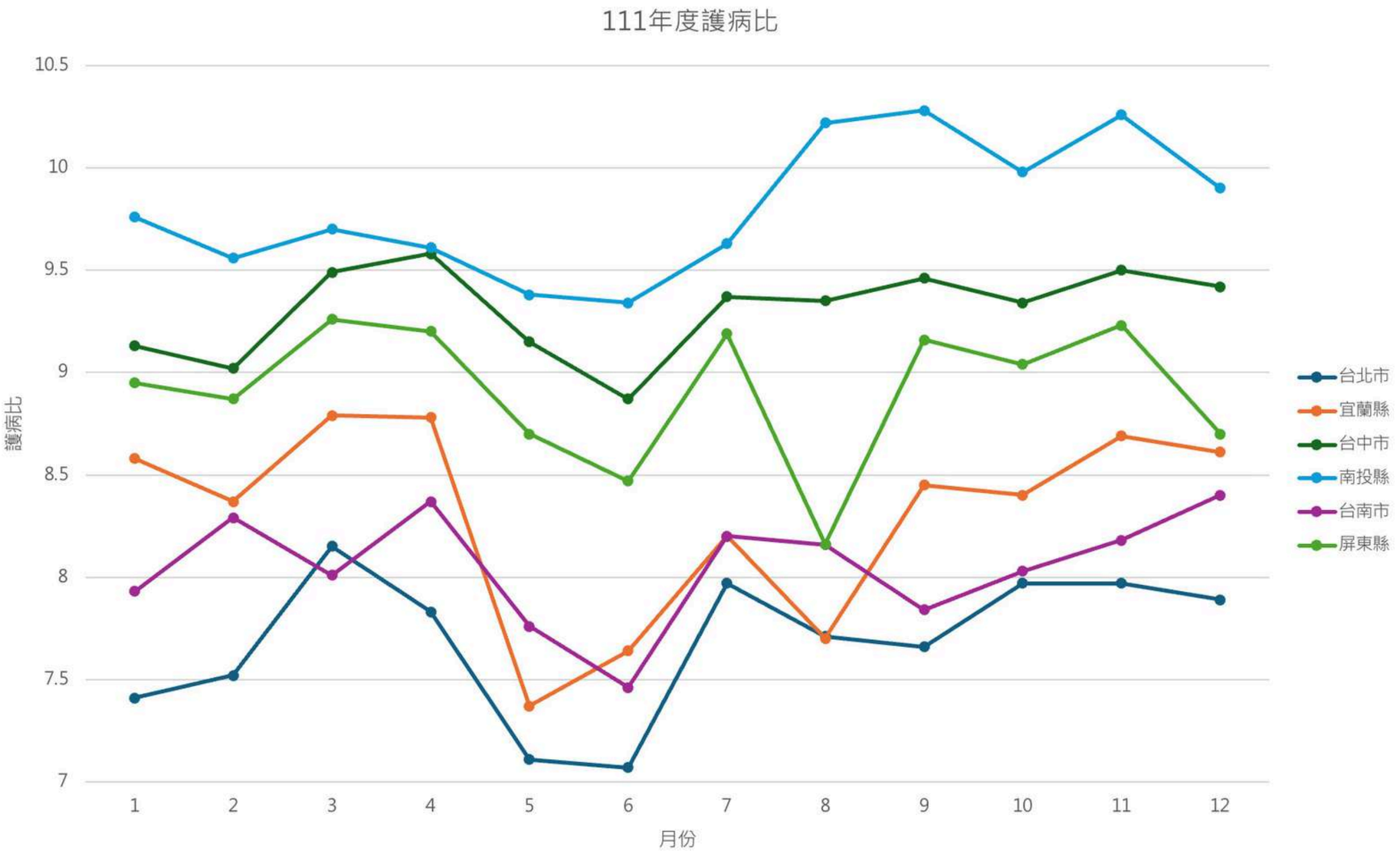
月份
城
鄉


111年度護病比
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南投縣
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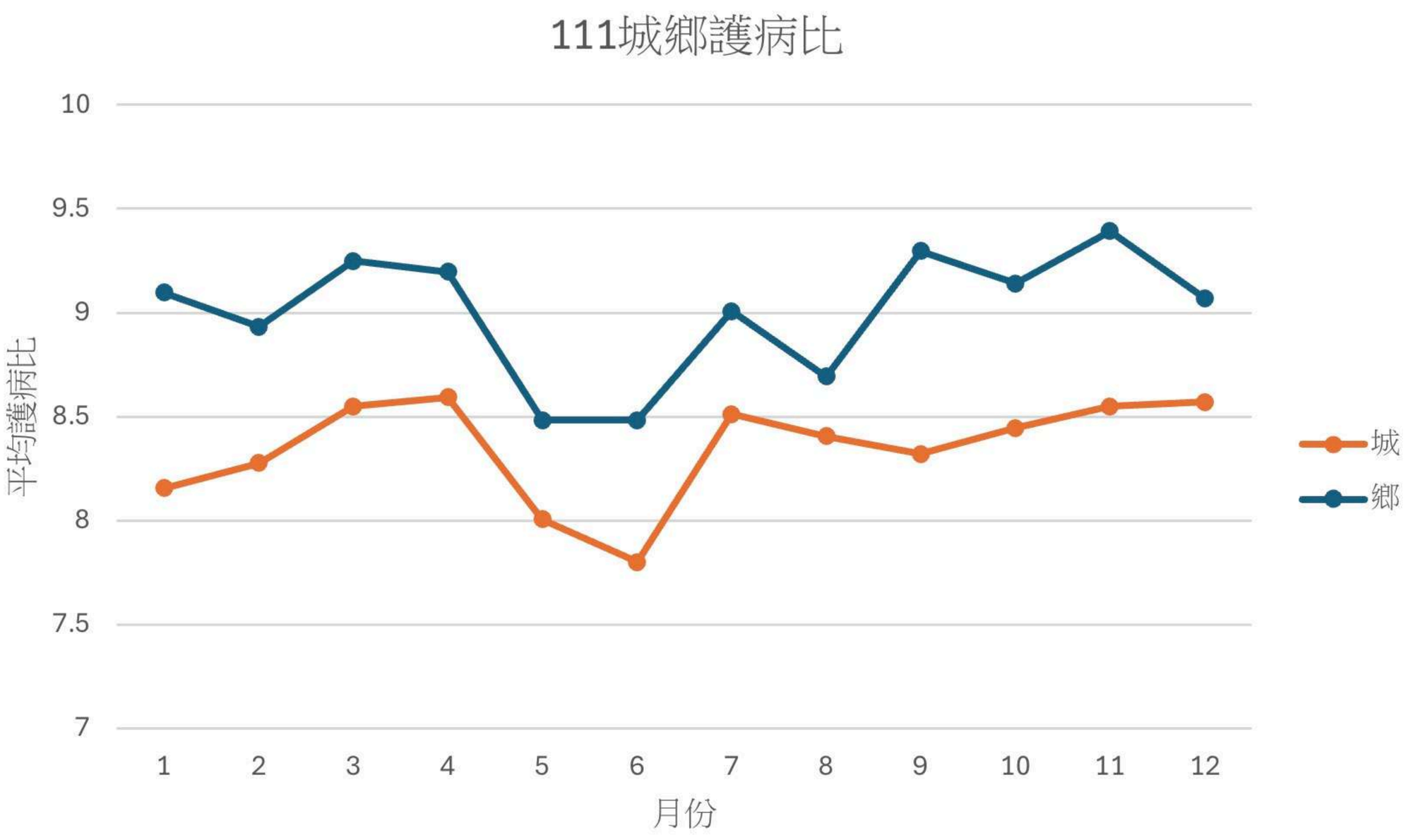
111城鄉護病比
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平均護病比
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12
月份
城
鄉

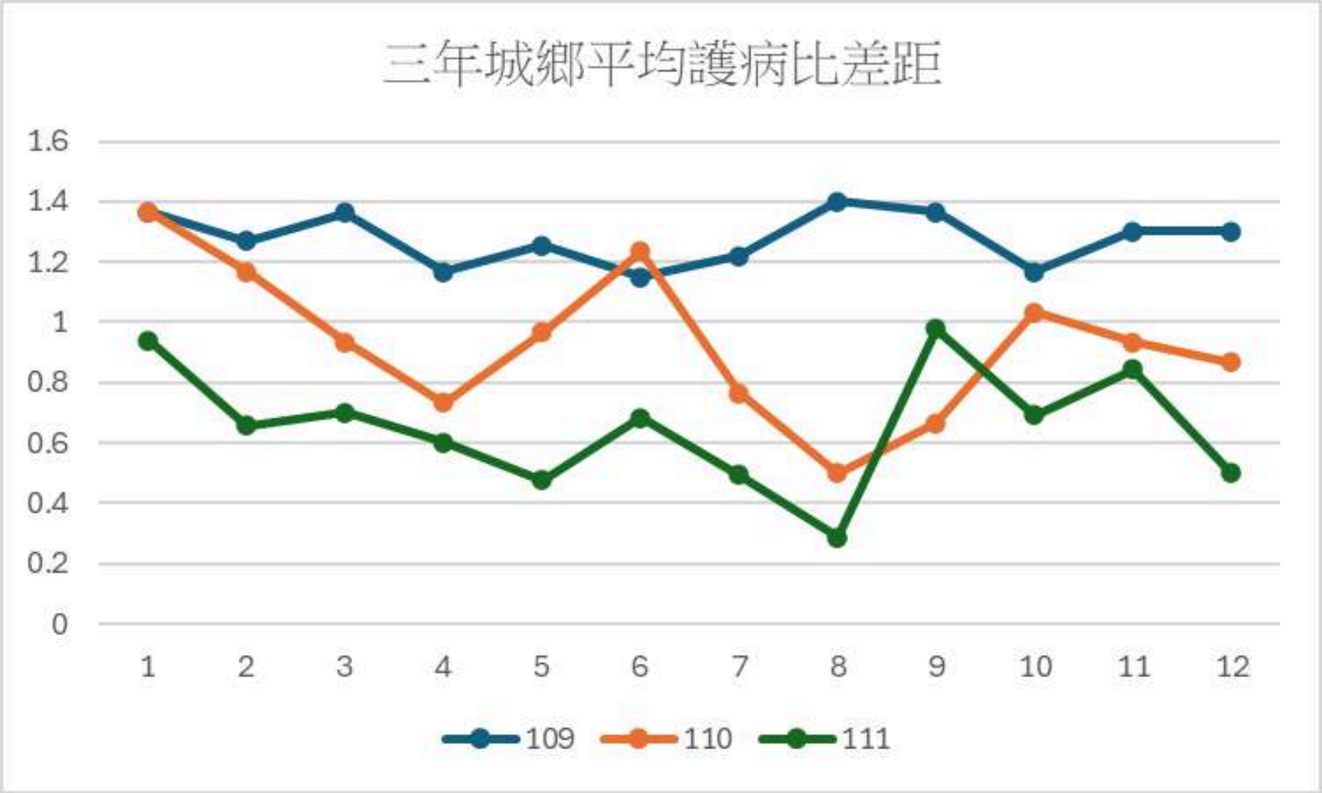
T-test data preparation

月份	台北	台中	台南	平均		宜蘭	南投	屏東	平均
1	8.1	8.9	7.8	8.2667		9.5	10.1	9.3	9.6333
2	8.6	9.7	7.8	8.7		9.1	10.3	10.2	9.8667
3	7.8	9.1	7.6	8.1667		8	9.8	9.5	9.1
4	8.6	9.7	8.5	8.9333		8.9	10.1	10	9.6667
5	8.5	9.7	8.6	8.9333		9.6	10.3	9.8	9.9
6	7.3	9.2	7.6	8.0333		8.7	9.8	9.3	9.2667
7	6.7	9.2	8.1	8		8	9.3	9	8.7667
8	7.3	9.3	8	8.2		7.9	9.2	9	8.7
9	7.8	9.3	8.2	8.4333		8.4	9.5	9.4	9.1
10	7.8	9.4	7.9	8.3667		8.6	10.1	9.5	9.4
11	8.1	9.2	7.9	8.4		8.6	9.8	9.6	9.3333
12	8.1	9.1	7.5	8.2333		8.5	9.4	9.4	9.1

	三年城鄉平均護病比差距		
月份	109	110	111
1	1.36667	1.3667	0.94
2	1.27	1.1667	0.65667
3	1.36333	0.93333	0.7
4	1.16667	0.7333	0.60333
5	1.25333	0.96667	0.47667
6	1.15	1.2333	0.68333
7	1.22	0.7667	0.49333
8	1.4	0.5	0.28667
9	1.36667	0.6667	0.97667
10	1.16667	1.0333	0.69333
11	1.3	0.9333	0.84333
12	1.3	0.8667	0.5

T-Test in Excel

	三年城鄉平均護病比差距		
月份	109	110	111
1	1.36667	1.3667	0.94
2	1.27	1.1667	0.65667
3	1.36333	0.93333	0.7
4	1.16667	0.7333	0.60333
5	1.25333	0.96667	0.47667
6	1.15	1.2333	0.68333
7	1.22	0.7667	0.49333
8	1.4	0.5	0.28667
9	1.36667	0.6667	0.97667
10	1.16667	1.0333	0.69333
11	1.3	0.9333	0.84333
12	1.3	0.8667	0.5



T-test variable characterization

mu = mu0	npr_aver_dif
The presumed population mean is 1.	The data sample designated for testing. (NPR represents the nurse-to-patient ratio)
alternative = c("fewer")	conf.int=0.95
It delineates the directionality of the test; "less" indicates whether the sample mean of the hypothesis is inferior to the population mean of the hypothesis.	The designated confidence level is 95%.

The average number of mothers is 1.

The internationally acknowledged ideal nurse-to-patient ratio is 1:6. In 2015, my country's evaluation standard indicated an average daily nurse-to-patient ratio of 1:9 for medical centers and 1:12 for regional hospitals.

Research indicates that when the nurse-to-patient ratio surpasses 1:6, the mortality rate rises by 7% for each additional patient assigned to a nurse.

Reference resource:

https://www.cw.com.tw/article/5077550

T-verification program code

```
exnpr_aver_dif <- c(1.366666667,1.27,1.363333333,1.166666667,
1.253333333,1.15,1.22,1.4,1.366666667,1.166666667,1.3,1.3,
1.366666667,1.166666667,0.933333333,0.733333333,0.966666667,
1.233333333,0.766666667,0.5,0.666666667,1.033333333,0.933333333,
0.866666667,0.94,0.656667,0.7,0.603333,0.476667,0.683333,
0.493333,0.286667,0.976667,0.693333,0.843333,0.5)
mu0 <- 1

t.test(exnpr_aver_dif,mu = mu0,alternative = c("less"),conf.int = 0.95)
```

```
        One Sample t-test

data:  exnpr_aver_dif
t = -0.86843, df = 35, p-value = 0.1955
alternative hypothesis: true mean is less than 1
95 percent confidence interval:
     -Inf 1.043513
sample estimates:
mean of x
0.9539815
```

- H0 = Average urban-rural disparity in patient-to-patient ratio ≤ 1
- H1 = Average urban-rural disparity in patient-to-patient ratio exceeding 1

Based on the test results, the P-value of 0.1955 exceeds the significance level of α = 0.05, leading us to accept (not reject) the null hypothesis (H0).

This suggests that there is no substantial difference in the nurse-to-patient ratio between urban and rural areas.

ANOVA Data Preparation

Data regarding the nurse-to-patient ratio in the six major cities in 2011

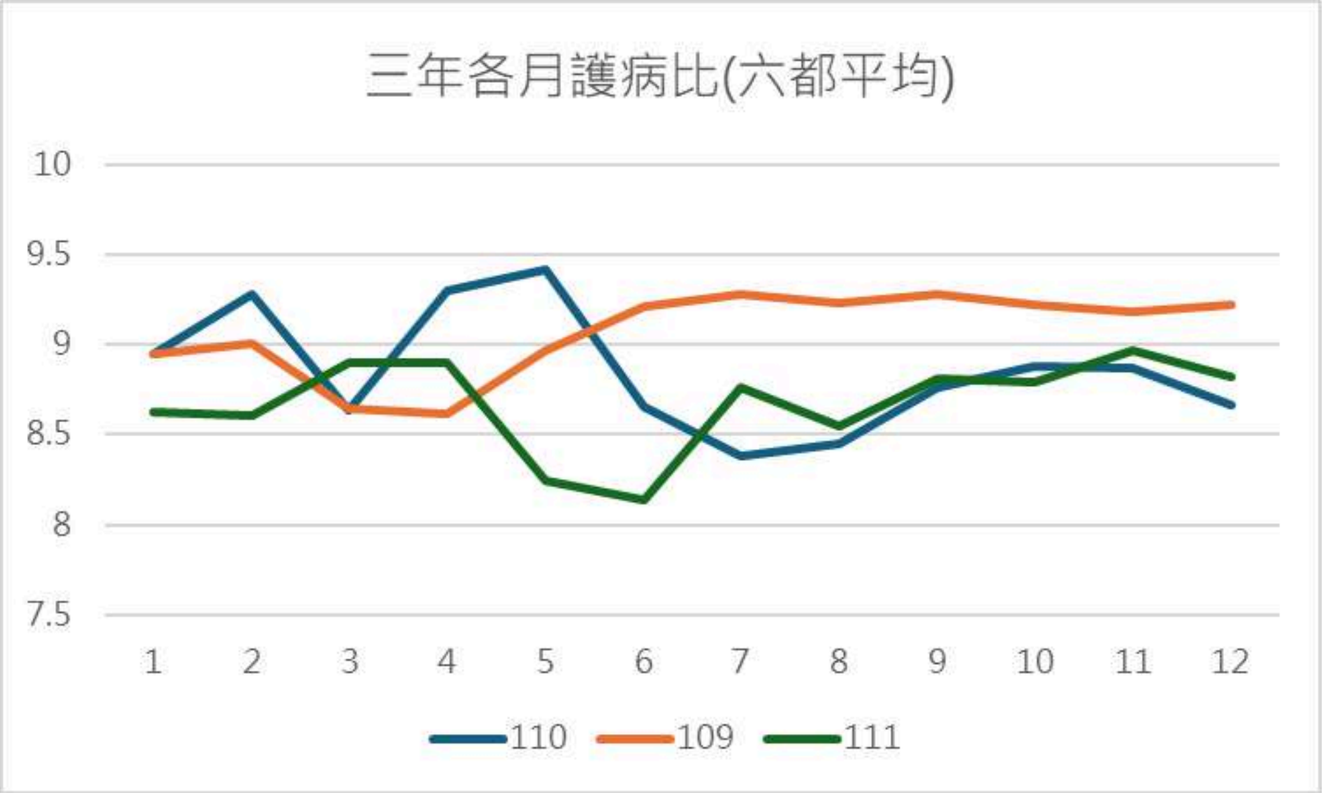
月份	台北	台中	台南	平均		宜蘭	南投	屏東	平均
1	8.1	8.9	7.8	8.2667		9.5	10.1	9.3	9.6333
2	8.6	9.7	7.8	8.7		9.1	10.3	10.2	9.8667
3	7.8	9.1	7.6	8.1667		8	9.8	9.5	9.1
4	8.6	9.7	8.5	8.9333		8.9	10.1	10	9.6667
5	8.5	9.7	8.6	8.9333		9.6	10.3	9.8	9.9
6	7.3	9.2	7.6	8.0333		8.7	9.8	9.3	9.2667
7	6.7	9.2	8.1	8		8	9.3	9	8.7667
8	7.3	9.3	8	8.2		7.9	9.2	9	8.7
9	7.8	9.3	8.2	8.4333		8.4	9.5	9.4	9.1
10	7.8	9.4	7.9	8.3667		8.6	10.1	9.5	9.4
11	8.1	9.2	7.9	8.4		8.6	9.8	9.6	9.3333
12	8.1	9.1	7.5	8.2333		8.5	9.4	9.4	9.1

Compute the mean.

	各月平均護病比(六都)		
月份	110	109	111
1	8.95	8.95	8.626667
2	9.28333	9.001667	8.605
3	8.63333	8.648333	8.9
4	9.3	8.616667	8.895
5	9.41667	8.963333	8.245
6	8.65	9.208333	8.141667
7	8.38333	9.276667	8.76
8	8.45	9.233333	8.55
9	8.76667	9.283333	8.808333
10	8.88333	9.216667	8.793333
11	8.86667	9.183333	8.971667
12	8.66667	9.216667	8.82

ANOVA in Excel

	各月平均護病比(六都)		
月份	110	109	111
1	8.95	8.95	8.626667
2	9.28333	9.001667	8.605
3	8.63333	8.648333	8.9
4	9.3	8.616667	8.895
5	9.41667	8.963333	8.245
6	8.65	9.208333	8.141667
7	8.38333	9.276667	8.76
8	8.45	9.233333	8.55
9	8.76667	9.283333	8.808333
10	8.88333	9.216667	8.793333
11	8.86667	9.183333	8.971667
12	8.66667	9.216667	8.82



ANOVA verification variable characterization

X	trt	npr
The dependent variable represents a quantified measurement. In this instance, it refers to the average nurse-to-patient ratio for each month across the six major cities.	The independent variable is employed to analyze measured values, with the objective of examining the impact of various classification treatments on the dependent variable. This example utilizes data segmented over three years pertaining to the pandemic.	A data frame comprising measurements and processing classifications.
npr.anova	anova.tab	TukeyHSD
Utilized for variance analysis	Utilized to create a variance analysis table.	Utilized for various comparative analyses.

ANOVA script

```
rm(list=ls())

######## function of anova table ############################

anova.tab <- function(fm){S
  tab <- summary(fm)
  k   <- length(tab[[1]])-2
  temp <- c(sum(tab[[1]][,1]),sum(tab[[1]][,2]),rep(NA,k))
  tab[[1]]["Total",]<-temp
  tab
}

#############################################################


npr <-
data.frame(X=c(8.95,9.283333333,8.633333333,9.3,9.416666667,8.65,8.3
95,9.001666667,8.648333333,8.616666667,8.963333333,9.208333333,9.276
67,8.626666667,8.605,8.9,8.895,8.245,8.141666667,8.76,8.55,8.8083333
),trt=gl(3,12,labels=c("110","109","111")))  ### 讀進資料 ###

npr.anova <- aov(X~trt, data=npr)

anova.tab(npr.anova)    ### 製作變異數分析表  ###

TukeyHSD(aov(X~trt,npr))   ### 利用 Tukey進行多重比較 ###
```

ANOVA script

Dependent variable:

- Average nurse-to-patient ratio per month (six major cities)

Independent variable:

```
            Df Sum Sq Mean Sq F value  Pr(>F)
trt          2  0.916  0.4578   5.877 0.00656 **
Residuals   33  2.571  0.0779
Total       35  3.486
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> TukeyHSD(aov(X~trt,npr))   ### 利用 Tukey進行多重比較 ###
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = X ~ trt, data = npr)

$trt
               diff         lwr        upr     p adj
109-110  0.2123611 -0.06723051  0.4919527 0.1652313
111-110 -0.1777778 -0.45736940  0.1018138 0.2768539
111-109 -0.3901389 -0.66973051 -0.1105473 0.0046242
```

Hypothesis test: H0: μA = μB = μC are equivalent.

H0: The severity of the COVID-19 pandemic will not influence the patient-to-career ratio.

•H1: μA, μB, and μC are not all identical.

H1: The impact of the COVID-19 pandemic will influence the nurse-to-patient ratio.

A. Monthly nurse-to-patient ratio in the six largest cities in 2020

B. Monthly nurse-to-patient ratio in the six largest cities in 2021 (epidemic year)

C. Monthly nurse-to-patient ratio in the six largest cities in 2022

ANOVA script

```
            Df Sum Sq Mean Sq F value  Pr(>F)
trt          2  0.916  0.4578   5.877 0.00656 **
Residuals   33  2.571  0.0779
Total       35  3.486
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> TukeyHSD(aov(X~trt,npr))    ### 利用 Tukey進行多重比較 ###
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = X ~ trt, data = npr)

$trt
              diff         lwr        upr     p adj
109-110  0.2123611 -0.06723051  0.4919527 0.1652313
111-110 -0.1777778 -0.45736940  0.1018138 0.2768539
111-109 -0.3901389 -0.66973051 -0.1105473 0.0046242
```

- 109-110 P-value = 0.1652313 > α = 0.05:
  The nurse-to-disease ratio did not exhibit a significant difference between 2010 and 2011.

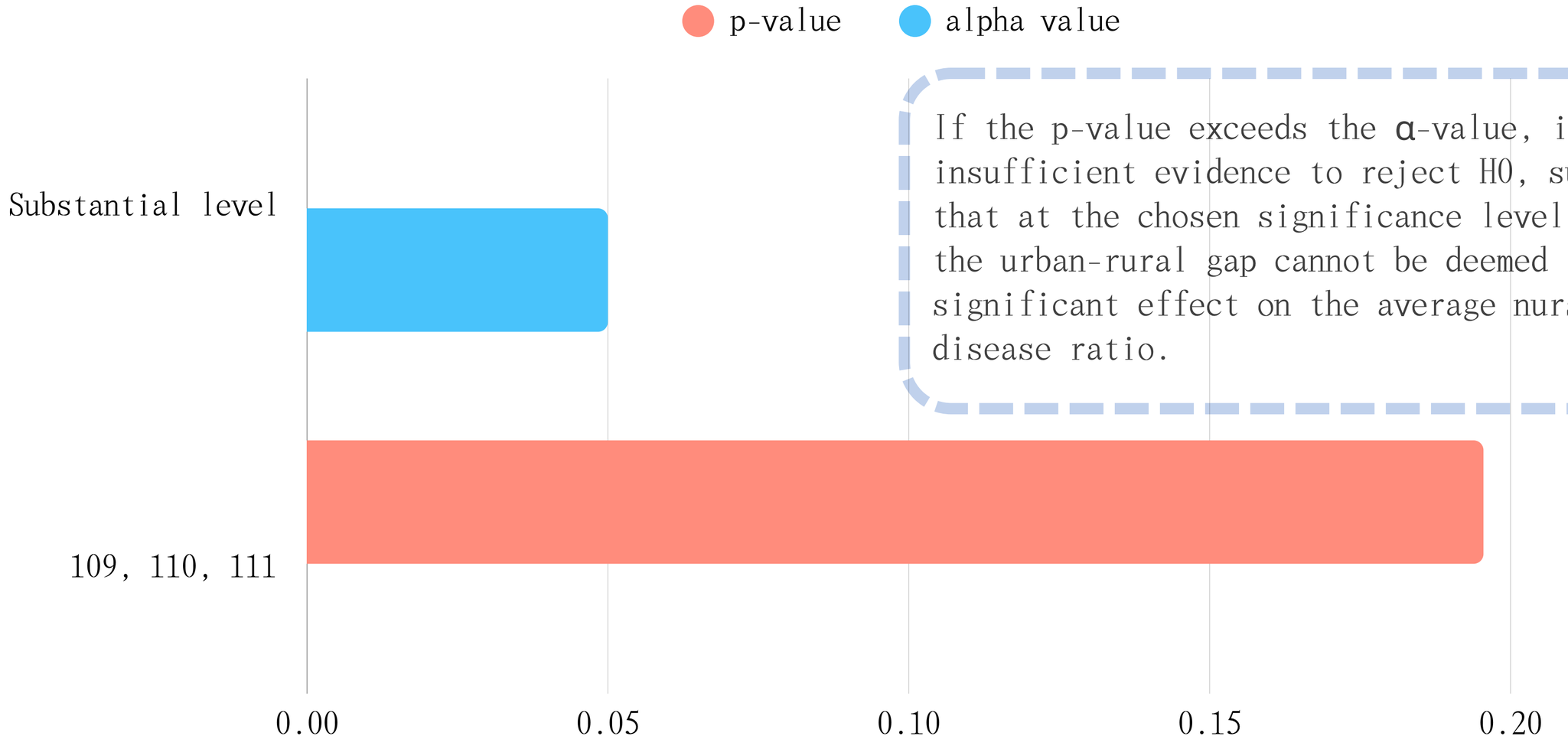
- 111-110 P-value = 0.2768539 > α = 0.05:
  The nurse-to-disease ratio did not exhibit a significant difference between 111 and 110 years.

- 111-109 P-value = 0.0046242 < α = 0.05:
  The nurse-to-disease ratio exhibited a significant variation between 2011 and 2019.
  This suggests that the nurse-to-disease ratio in 2011 was markedly distinct from that in 2019.

Conclusion

Employing T-TEST to evaluate hypothesis 1

During the three years of the COVID-19 pandemic, is the average patient-to-patient ratio associated with the urban-rural divide?

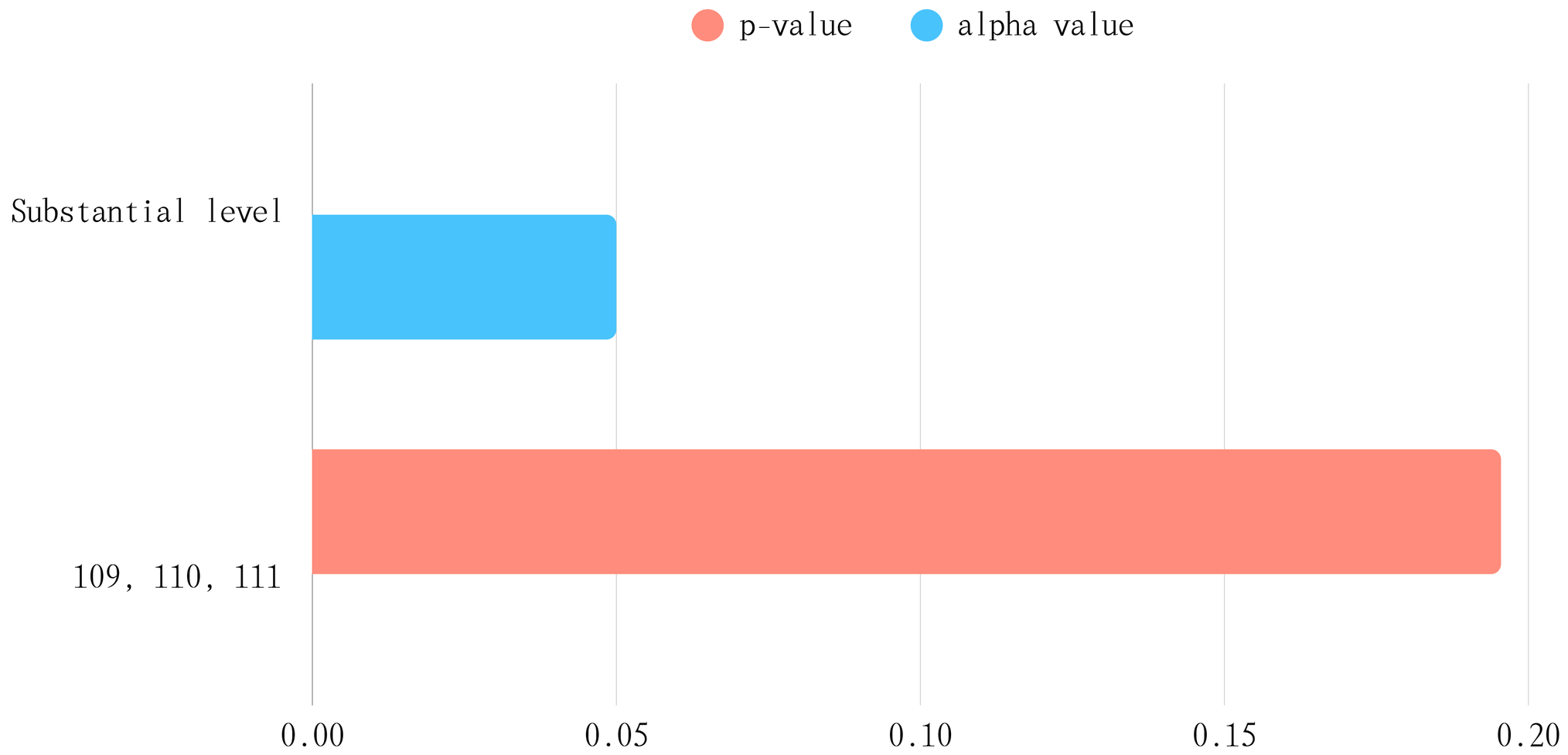


If the p-value exceeds the α-value, it indicates insufficient evidence to reject H0, suggesting that at the chosen significance level of α=0.05, the urban-rural gap cannot be deemed to have a significant effect on the average nurse-to-disease ratio.

Employing T-TEST to evaluate hypothesis 1

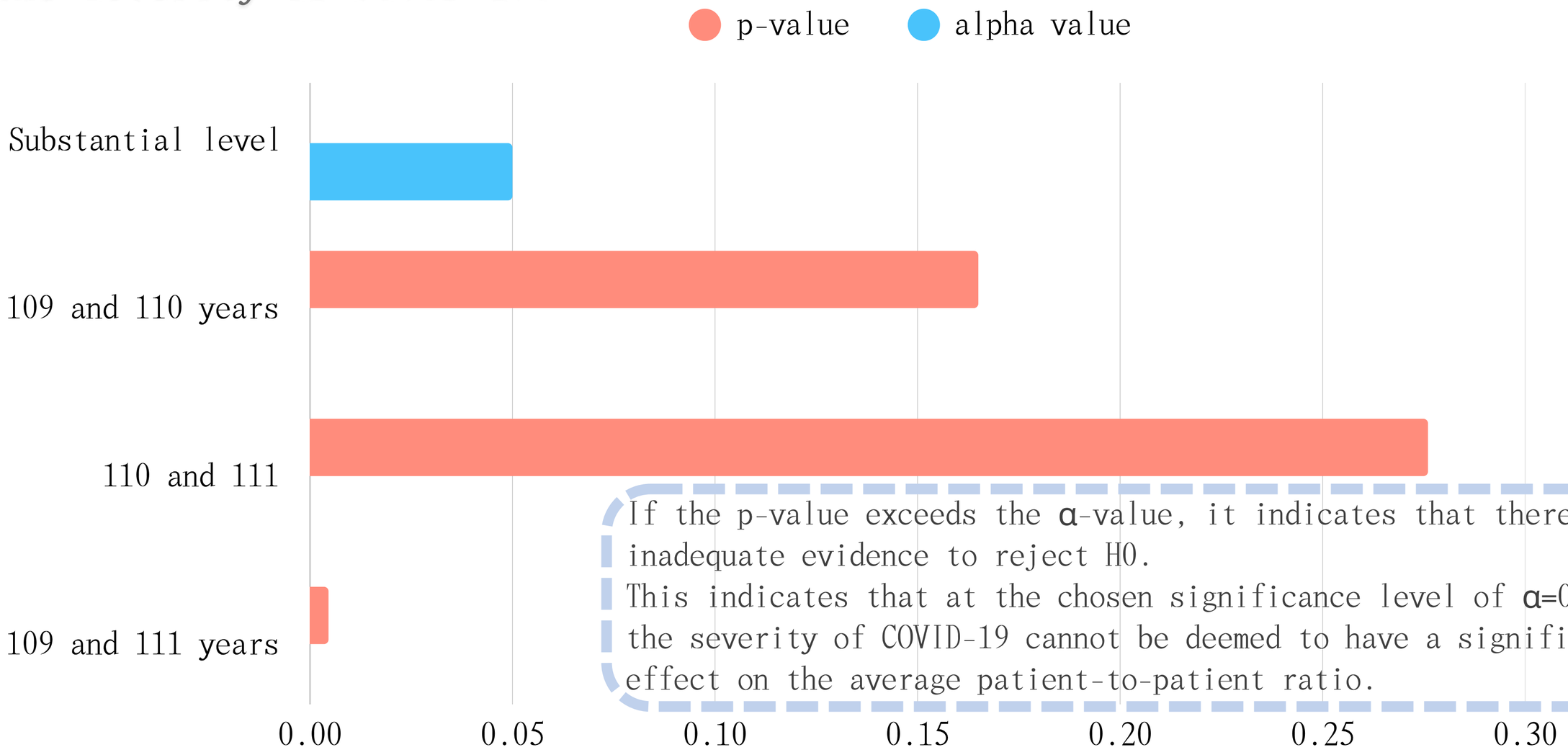
Inference: The average quality of nursing care is not the primary factor contributing to disparities in rural regions.

The urban-rural divide did not expand as a result of COVID-19, enabling non-urban regions to access medical care as well.



Hypothesis 2 was evaluated through ANOVA.

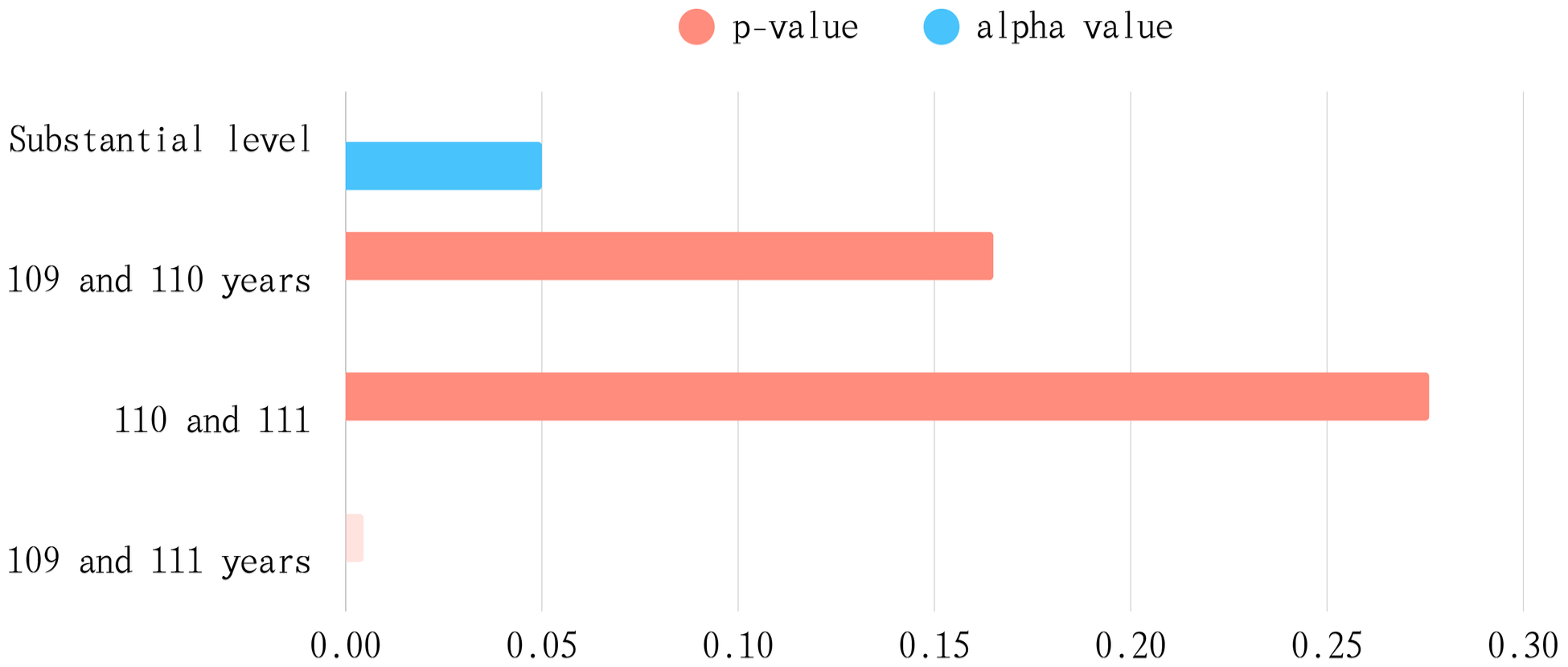
During the three years of the COVID-19 pandemic, what was the average nurse-to-patient ratio in the six major cities significantly impacted by the severity of COVID-19?



If the p-value exceeds the α-value, it indicates that there is inadequate evidence to reject H0.
This indicates that at the chosen significance level of α=0.05, the severity of COVID-19 cannot be deemed to have a significant effect on the average patient-to-patient ratio.

Employing ANOVA to evaluate hypothesis 2

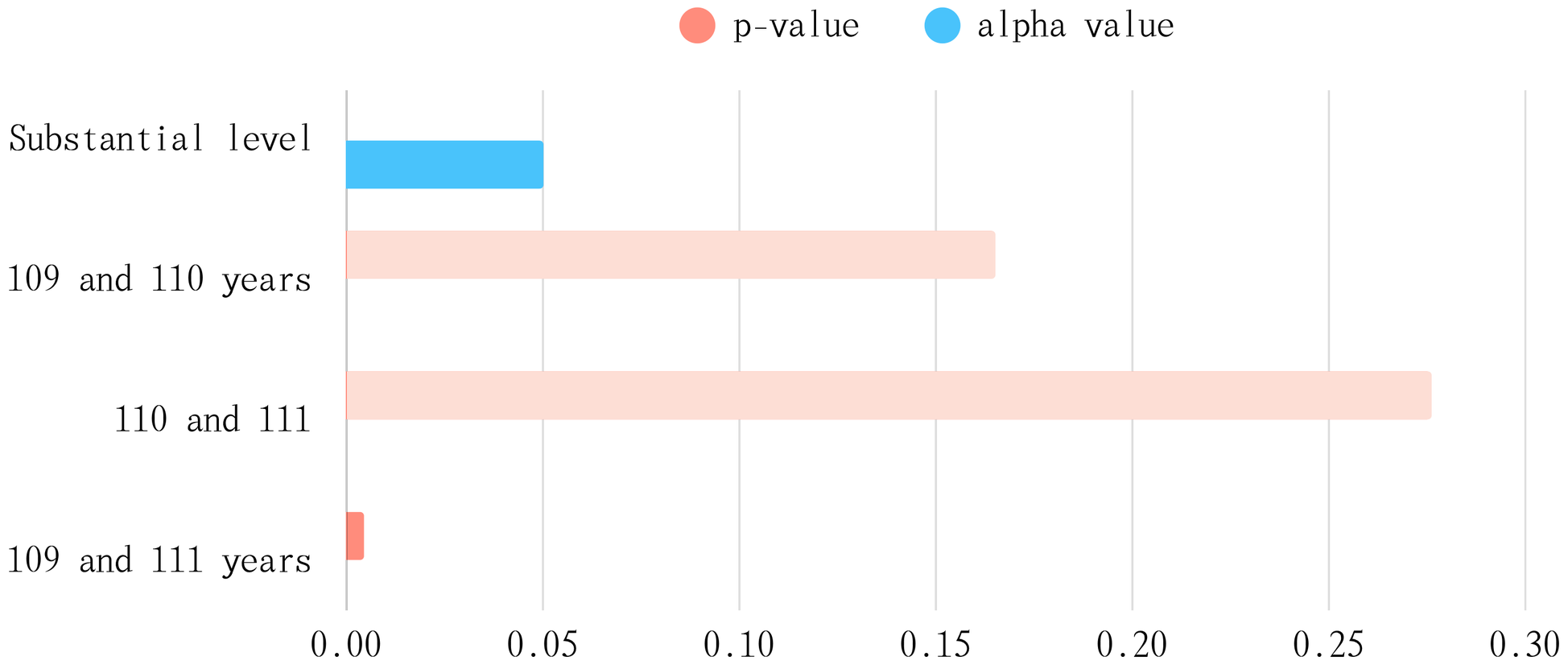
The average number of patients undergoing treatment did not vary significantly based on the severity of the epidemic.



- There was no notable difference in the average nurse-to-disease ratio between 2011 and 2010/2012/2012.
- Although 110 was the year marked by the most severe epidemic, the average patient-to-patient ratio remained relatively stable, according to the data.

Employing ANOVA to evaluate hypothesis 2

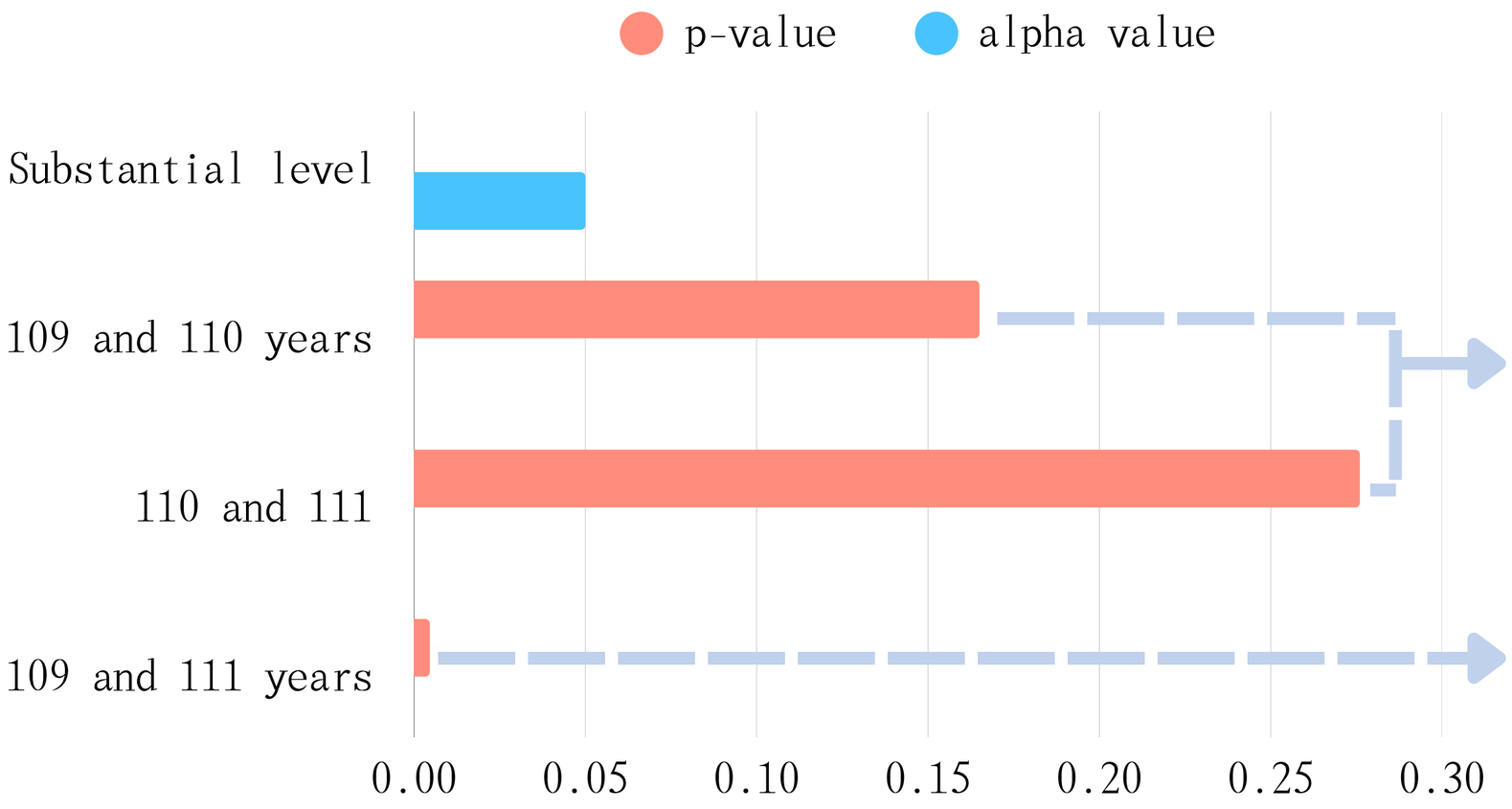
As the pandemic subsided, the average patient count exceeded that of the pandemic's onset.



2012 marked a year in which the pandemic subsided; however, the average patient-to-patient ratio was considerably higher than in 2010, the year the pandemic commenced.

Employing ANOVA to evaluate hypothesis 2

This indicates that there may have been modifications to medical resources or alterations in policies.



It can be inferred that in 110 AD, in response to the severe epidemic, the medical system implemented corresponding adjustments, ensuring that the nurse-to-patient ratio did not increase significantly.

Inferences can be made concerning the long-term effects of the pandemic on the healthcare system.
This led to the patient ratio remaining elevated even after the pandemic subsided compared to its level at the onset of the pandemic.

Task Allocation

Work Assignment Schedule

member	Job responsibilities
411570032 Xu Shaoyun	Variable description, statistical analysis -- ANOVA examination
411570068 Lin Yixiu	Variable description, statistical analysis -- T-test
411570111 Weng Zhenwei	Objectives, Hypothesis Formulation, Summary
411570135 Lin Wanzhen	Motivation, Synopsis
411570147 Wang Caixun	Organizing data sources and creating PowerPoint presentations.
411570202 Li Chenxuan	Data collection and statistical analysis -- Calculation of nurse-to-disease ratio